**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 081 · Syntax and Logic Errors

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why must a trace table follow executed assignments in order?

Q2. Why open a file before reading it?
A. To create a file handle
B. To delete its contents
C. To encrypt every line
D. To print it automatically

Q3. “Reusing one short password everywhere improves account security.” Correct this. Explain why.

Q4. A file is split into fragments on an HDD. “Defragmentation speeds up every SSD by moving its read head.” Correct this. When does defragmentation help?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

Syntax error — Statement breaking the grammar of a language.

Logic error — Program runs but produces a wrong result.

Trace table — A record of variable values as an algorithm executes.

Debugging — Finding and correcting program faults.

Retesting — Repeating a test after correcting a bug.

Boundary test — Testing at and immediately either side of input limits.

### New learning

Syntax errors: A misspelled IF keyword or missing ENDIF violates grammar. In a runnable implementation it may prevent the code from executing.

Logic errors: A program may run but give the wrong result, for example using addition when the task needs multiplication.

Find the first incorrect step: Trace variables as instructions execute. Compare expected and observed values to pinpoint the first incorrect result.

Fix and retest: Correct the faulty instruction then repeat the failed case and nearby boundary cases to check the effect.

### Worked example

A ticketing program calculates total = price + quantity when it should calculate cost for multiple tickets. With price = 4 and quantity = 3, identify the error, correct it and explain the result. [4 marks]

1. The expected cost is three tickets priced at £4 = £12 [1].

2. The program uses addition, producing 4 + 3 = 7 [1].

3. Replace + with multiplication * [1].

4. After the correction 4 * 3 gives 12 [1].

## We Do · Practise with guidance

### Guided 1

A sports club program should find four times a membership fee, but adds 4 to it. With fee = 6, explain the logic error and fix. [4 marks]

Worked method: Addition would calculate 6+4=10 [1], rather than 24 [1]. Change operator to multiplication [1], producing 6*4=24 [1].





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show the first incorrect value [1]. Repeat failed test after correction [1] and test another normal/boundary case to avoid regressions [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Each variable value depends on the state produced by earlier executed instructions.

Q2. A — To create a file handle

Q3. They make guessing harder and prevent reuse of a compromised password across accounts.

Q4. Defragmentation brings file fragments together on an HDD, reducing read-head movement. It rearranges data; it does not encrypt or compress it.

### I Do / We Do pages · Guided practice

Q1. Addition would calculate 6+4=10 [1], rather than 24 [1]. Change operator to multiplication [1], producing 6*4=24 [1].

Q2. Missing ENDIF leaves an IF block unclosed [1]. Add ENDIF after the appropriate branch to complete the block [1].

Q3. Repeat the originally failing test [1] to confirm the bug is fixed [1]. Also test a boundary or another input [1] to check the change did not cause another failure [1].

### You Do pages · Code First

```
bits = 32
bytes = bits DIV 8
print(bytes)
```

Expected result: 4

### You Do pages · Independent questions

Q1. Syntax: invalid language grammar [1]; logic: program runs but produces incorrect results [1].

Q2. Adding gives 8 [1] but expected price is 15 [1]; use multiplication [1], giving 3*5=15 [1].

Q3. A syntax error breaks language rules [1] and should be corrected according to diagnostic feedback [1]. Logic errors produce wrong results [1] and trace tables



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### Guided 2

An IF block is missing ENDIF in an example. Explain the syntax error and how to correct it. [2 marks]

Worked method: Missing ENDIF leaves an IF block unclosed [1]. Add ENDIF after the appropriate branch to complete the block [1].

### Guided 3

A pupil corrects a program but tests it only once. Explain two improvements to the testing process. [4 marks]

Worked method: Repeat the originally failing test [1] to confirm the bug is fixed [1]. Also test a boundary or another input [1] to check the change did not cause another failure [1].

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## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Fix the calculation that converts 32 bits into bytes.

```
bits = 32
bytes = bits * 8
print(bytes)
```

iterate

### Question 1

State the difference between a syntax error and a logic error. [2 marks] Source: Original OCR J277-style practice question.

### Question 2

A fee calculator adds 5 to the number of tickets instead of multiplying by 5. With 3 tickets, identify the error and correct the calculation. [4 marks] Source: Original OCR J277-style practice question.

### Question 3

Explain how tracing and retesting can identify, fix and confirm both syntax and logic errors in a program. [6 marks] Source: Original OCR J277-style practice question.



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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

## Exit Ticket · Answer before checking

A logic error can cause an incorrect output even when the program runs.